

Problem

A circuit has $R_1 = 3\ \Omega$, $R_2 = 10\ \Omega$, $L = 2\text{ H}$, and $C = 1/10\text{ F}$. After the circuit is magnitude-scaled by 100 and frequency-scaled by 10^6 , find the new values of the circuit elements.

Step-by-step solution

Step 1 of 1

$$R_1' = K_m R_1 = 3 \times 100 = 300\ \Omega$$

$$R_2' = K_m R_2 = 10 \times 100 = 1\text{ K}\Omega$$

$$L' = \frac{K_m}{K_f} L = \left(\frac{10^2}{10^6} \right) (2) = 200\ \mu\text{H}$$

$$C' = \frac{C}{K_m K_f} = \frac{1}{10^8} = 10\text{ nF}$$